THE BONDI DIPOLE IN FULL NUMERICAL RELATIVITY: A SELF-ACCELERATING POSITIVE–NEGATIVE MASS BINARY

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Bondi observed in 1957 that bodies of opposite *active* gravitational mass self-accelerate: the negative one chases the positive one it repels, and the pair runs off together. We evolve this "Bondi dipole" in 3+1 numerical relativity: two complex scalars with identical Klein–Gordon dynamics, of which only the phantom enters Einstein's equations with a minus sign—inertial and passive masses positive, active mass negative. Across twenty uniform-grid evolutions at three resolutions, the mixed pair—and only the mixed pair—runs away: released at rest at separation $d_0 = 12$, its barycentre drifts $+1.64$ by $t = 60$, 46 and 142 times the same-sign controls, stable to 6.3% under refinement while the controls converge away. The drift reverses under mirroring (4.7%), scales with the phantom's active mass (0.7–3.9%), and the $\ell = 2$ Weyl amplitude tracks the runaway speed at correlation 0.996; the signed sector momenta cancel to a residual matching $M_+v_+ + M_-v_-$ to $\sim$10%. The phantom star is, to our knowledge, the first stable asymptotically flat body of negative ADM mass evolved in numerical relativity.

I. INTRODUCTION

Bondi [1] gave negative mass its modern treatment, distinguishing inertial, passive and active gravitational mass and noting the consequence of mixing signs: a negative-mass body is *attracted* by a positive one, the positive body is *repelled* by the negative one, and the pair runs off together (Fig. 1)—momentum conserved, because the negative body's contribution is negative. Bonnor and Swaminarayan [2] found exact solutions for uniformly accelerated opposite-mass pairs, later generalized in Ref. [3]; Forward [5] showed "in tedious detail" that the self-accelerating pair violates neither momentum nor energy conservation, for equal and unequal mass magnitudes alike, so objections to negative mass must be sought elsewhere [6, 7].

Why put this in full numerical relativity? Because nothing in Einstein's equations forces mass to be positive—positivity is a theorem only once energy conditions are imposed [26, 27]—and modern physics keeps producing sources that strain those conditions: Casimir energies, quantum fields in curved spacetime, phantom dark energy [8]. Every traversable wormhole [9] and warp spacetime [10] needs matter of exactly this kind, and the standing objection to all of them is Bondi's runaway itself—matter that accelerates for free looks like a perpetual-motion machine [6, 7]. That objection has only ever been examined for point particles and prescribed metrics; the nonlinear evolutions that exist for energy-condition-violating matter—ghost-scalar wormholes in spherical symmetry [14, 29], static phantom-field stars [30]—concern single objects. Whether the runaway survives full nonlinear gravity—finite bodies, backreaction, radiation—is a concrete, checkable question, and we are not aware of a previous 3+1 evolution of a gravitationally bound positive–negative active-mass pair with dynamical, constraint-solved matter.

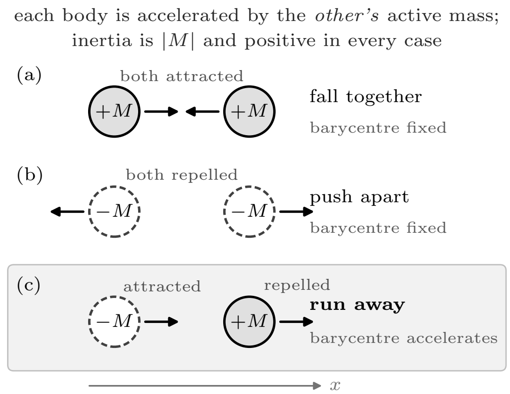

FIG. 1. Bondi's sign rules for two bodies released at rest, in the style of Forward's analysis [5]. Every body has positive inertial and passive mass; only the *active* (source) mass carries a sign—negative for the dashed bodies. (a) Two positive-mass stars attract and fall together; (b) two negative-mass stars repel and push apart; in both the forces oppose and the pair barycentre stays put. (c) The mixed pair—the Bondi dipole: both forces point the *same* way, so the pair barycentre accelerates while the system carries zero total momentum in the signed bookkeeping. Each row is a run of the matrix in Table I.

The question has acquired observational stakes: negative-mass binaries have been proposed as sources of anomalous gravitational-wave signatures—anti-chirps, dispersal, runaway motion—turning Bondi's paradox into an exclusion channel for real detectors [38], while Nojiri and Odintsov [12] find that a bound positive–negative system *can* form for a compact mass immersed in a cosmological fluid with negative cosmological constant. Our asymptotically flat, isolated pair is the complementary case, and it does not bind—it runs away.

We build a minimal field-theoretic realization of

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Bondi's sign structure (Sec. II); solve dressed soliton stars of both signs as constraint-satisfying initial data (Sec. III); run a falsifiable matrix at three resolutions per cell (Sec. IV); report the measurement, in which the mixed pair alone self-accelerates with a grid-stable drift while the controls converge away (Sec. V); read the geometric witness in the $\ell = 2$ Weyl amplitude, with a wave-zone extraction on a doubled box (Sec. VI); and account for the diagnostic biases that could mimic the effect (Sec. VII). Every evolution ran on a single GPU with `GRTeclyn` [23], the AMReX-based [25] GPU port of `GRChombo` [22]; with the wormhole campaign of Ref. [39] these are, to our knowledge, the first evolutions of energy-condition-violating matter in GPU-native numerical relativity. Throughout, $G = c = 1$, the scalar mass sets the unit ($m = 1$), drifts along the pair axis are $\Delta x = x(t) - x(0)$, and d is the instantaneous barycentre separation.

II. THE BICOMPLEX SCALAR MODEL

The matter must keep canonical *dynamics*—positive inertial and passive mass—while only its gravitational *source* changes sign; it must form stable, localized stars; and the negative-mass star must be held together by something other than gravity, since its own gravity pushes it apart. Each obvious candidate fails one of these: the negative-mass Schwarzschild spacetime is a naked singularity with exponentially growing smooth linear perturbations [13] and contains no matter from which to build initial data; a negative-energy fluid has no equilibrium to sit in, Bonnor's static spheres [4] demanding an inverted interior with no non-gravitational restoring force; and a ghost scalar reverses its field dynamics along with its source, flipping inertial and passive masses too (Forward's all-negative matter [5], not the signature under test), and such fields collapse or explode in evolution [14, 29, 39].

What passes is a *soliton*: matter self-bound by its own interactions, for which gravity is a dressing rather than the binding agent. We take two independent complex scalars, Φ_+ (*canonical*) and Φ_- (*phantom*), on one spacetime, sharing the standard sextic potential of Q-ball and solitonic boson-star studies [15–17],

$$V(|\Phi|^2) = \tfrac{1}{2}m^2|\Phi|^2 - \tfrac{1}{4}\lambda|\Phi|^4 + \tfrac{1}{6}\mu|\Phi|^6, \tag{1}$$

with $\lambda, \mu > 0$: the mass term makes the field oscillate, the attractive quartic lets it clump, the repulsive sextic stops the clumping short of collapse. Sharing one potential makes the matter physics of the two sectors *identical*, so the single sign introduced next is the only possible cause of any asymmetry the simulations show.

That sign lives in one place—the Einstein equations are sourced by the *signed* sum of two canonical stress tensors,

$$G_{\mu\nu} = 8\pi\left(T_{\mu\nu}[\Phi_+] - T_{\mu\nu}[\Phi_-]\right), \tag{2}$$

while the field equations of *both* sectors keep their canonical form,

$$\Box\,\Phi_\pm = 2\,\frac{\partial V}{\partial|\Phi|^2}\,\Phi_\pm, \tag{3}$$

the factor 2 fixed by the $\frac{1}{2}$-normalized kinetic term. In 3+1 language the sources seen by normal observers accumulate per sector with a sign $\epsilon_\pm = \pm 1$,

$$\rho = \sum_{s=\pm}\epsilon_s\,\rho[\Phi_s], \qquad S_i = \sum_{s=\pm}\epsilon_s\,S_i[\Phi_s], \tag{4}$$

and likewise for S_{ij}, each bracket being the standard scalar expression. The model derives from

$$S = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi} + \mathcal{L}[\Phi_+] - \mathcal{L}[\Phi_-]\right], \tag{5}$$

with $\mathcal{L}[\Phi] = -\frac{1}{2}\nabla_\mu\Phi^*\nabla^\mu\Phi - V(|\Phi|^2)$: a sector's overall Lagrangian sign cancels from its Euler–Lagrange equations—hence Eq. (3)—but survives the metric variation—hence Eq. (2). Bianchi consistency is automatic (each sector's stress tensor is separately conserved on shell, so the signed sum is too); well-posedness is inherited (the sector sign enters CCZ4 only through algebraic source terms); and the mass signature is Bondi's: the phantom *moves* canonically—positive inertial and passive mass—but *gravitates* with the minus sign of Eq. (2), so its active mass, and its ADM mass, are negative. Place one star of each kind side by side and both accelerations point from phantom toward canonical: Newton's third law "fails" in the precise sense that the two forces point the same way, and no conservation law breaks, because the conserved momentum belongs to the signed stress-energy of Eq. (4).

The phantom sector violates the null energy condition by construction, and quantum viability is not claimed—a sector entering gravity with negative energy admits graviton-mediated vacuum decay [31, 32]—so the model is a classical probe of *if such matter existed, what would gravity do*, in the spirit of phantom cosmology [8]. It is implemented in `GRTeclyn` [22, 23, 25], evolving CCZ4 [21] plus the two fields, with Eq. (4) accumulated star by star in the initial-data solver `GRTresna` [24] so that one constraint solve can carry mixed-sign matter.

III. INITIAL DATA: DRESSED SOLITONS

A. Couplings

A soliton of this theory oscillates coherently, $\Phi \propto e^{-i\omega t}$, and ω labels the solution family: $\omega = m$ is the free-field limit, $1 - \omega$ the binding energy per unit mass. The ratio λ^2/μ sets the lowest attainable frequency, $\omega_{\min} = \sqrt{1 - 3\lambda^2/16\mu}$ [16]; all runs use $\lambda^2/\mu = 4.8$, so $\omega_{\min} = 0.316$ and the working $\omega = 0.55$ sits well inside the range, at 45% binding. The overall coupling strength, $\lambda = 10240$ and $\mu \simeq 2.18 \times 10^7$ everywhere (amplitude $\simeq$0.019), is not a free choice: every more strongly gravitating combination produced stars that collapsed or dispersed.

B. The dressed-star problem

A flat-space soliton cannot simply be placed on an initial slice: the constraints are violated by a bare Q-ball, and once its own gravity acts the profile is no longer an eigenstate—seeded anyway, the star breathes, sheds and disperses. The star we need is solved with its gravity *on*: the boson-star problem [18–20] with a sextic self-interaction, posed for both source signs. With $\Phi = \varphi_0(r)e^{-i\omega t}$ and a conformally flat, maximally sliced metric $ds^2 = -\alpha^2 dt^2 + \psi^4 \delta_{ij} dx^i dx^j$, Eqs. (2)–(3) reduce to

$$\begin{aligned} \psi'' + \tfrac{2}{r}\psi' &= -2\pi\psi^5\rho, \\ \alpha'' + \left(\tfrac{2}{r} + \tfrac{2\psi'}{\psi}\right)\alpha' &= 4\pi\psi^4\alpha\,(\rho + S), \\ \varphi_0'' + \left(\tfrac{2}{r} + \tfrac{\alpha'}{\alpha} + \tfrac{2\psi'}{\psi}\right)\varphi_0' &= \psi^4\left(\frac{dV}{d\varphi_0} - \frac{\omega^2}{\alpha^2}\varphi_0\right), \end{aligned} \tag{6}$$

with $\rho = \frac{1}{2}\omega^2\varphi_0^2/\alpha^2 + \frac{1}{2}\psi^{-4}\varphi_0'^2 + V$ and $\rho + S = 2\omega^2\varphi_0^2/\alpha^2 - 2V$; here $dV/d\varphi_0 \equiv 2\varphi_0\,\partial V/\partial|\Phi|^2$. **The phantom star is obtained by flipping the sign of ρ and $\rho + S$ in the metric equations only**; the Klein–Gordon line keeps its canonical form, mirroring Eqs. (2)–(3). The result is a self-interaction-bound soliton with repulsive gravity: $\psi < 1$, $\alpha > 1$ in the core, and negative ADM mass.

We integrate Eqs. (6) by shooting *at fixed frequency*: heavy-branch sextic stars sit a fraction of a percent below the effective-potential top, so at fixed central amplitude the eigenvalue in ω is an exponentially thin needle and a shooter on ω lands on the light branch. We bisect the central amplitude φ_c at fixed integration-frame frequency and outer-iterate until ω/α_∞ matches the request, to $\sim 10^{-5}$.

Figure 2 shows both families. The reference pair uses the two $\omega = 0.550$ stars, canonical $M_{\rm ADM} = +0.063951$ and phantom $M_{\rm ADM} = -0.076962$: at equal frequency the phantom branch is heavier, so the reference pair is intrinsically unequal, $|M_-|/M_+ = 1.203$. For the equal-$|M|$ run the phantom is instead solved at $\omega = 0.56598$, where $M_{\rm ADM} = -0.063950$. Matching the *ADM* masses—not bare amplitudes or Noether charges—is what Bondi's point-mass limit dictates: each body's exterior field, and hence the force it exerts, is set by its ADM mass alone.

C. From radial profiles to three-dimensional data

The solved profiles are transferred onto the Cartesian grid, one star at each release position. Two details are

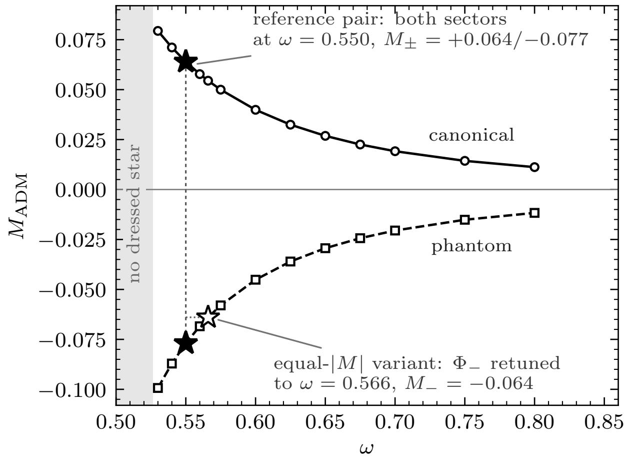

FIG. 2. ADM masses of the spherical stars against oscillation frequency, from Eqs. (6) at the working couplings. Lower ω means deeper binding and larger $|M_{\rm ADM}|$; in the shaded region the solver finds no dressed star. At equal frequency the phantom star is the heavier in magnitude (20.3% at $\omega = 0.550$), its repulsive self-gravity spreading it over more volume. Filled stars: the reference pair, both sectors at $\omega = 0.550$. Open star: the equal-$|M_{\rm ADM}|$ variant, phantom retuned to $\omega = 0.56598$ so that M_- matches the canonical mass to one part in 10^5.

essential. The seeded amplitude must be the star's own solved φ_c; any rescaling de-tunes the eigenstate. And a boson star is stationary but not static: its phase rotates, and the conjugate momentum is defined per unit *proper* time of the normal observers, $\Pi = \alpha^{-1}\partial_t\Phi$ for zero shift, so the harmonic phase seeds

$$\Pi_{\rm im} = -\frac{\omega}{\alpha(r)}\,\varphi_0(r); \tag{7}$$

a flat-space seed ($\alpha \equiv 1$) would start the star off its eigenstate wherever the lapse differs from one. Each star is seeded from its own table with its own frequency.

`GRTresna` then solves the constraints for the pair (CTTKHybrid scheme [34], sources summed with their signs star by star) to 0.1%; at an earlier 1% tolerance the leftover residual radiated away as a ring in χ that shook the stars' envelopes. Every production cell exits at Hamiltonian error $\leq$0.095% and momentum error $\leq$0.080% in seven iterations or fewer (Appendix A).

All pair runs seed the two lumps with $U(1)$ phases 0 and π. For the mixed pair this is inert (the sectors are independent fields, so their relative phase is pure gauge), but in the same-sector controls both lumps live in *one* complex field, where superposed boson-star data is known to inject phase-dependent artifacts that a constraint re-solve reduces but does not eliminate [33]. The full solve to 0.1% bounds this systematic; a phase-offset `PP` rerun is future work.

IV. THE CAMPAIGN

A. Numerics and the run matrix

Evolutions use `GRTeclyn` (one GPU per run, single rank): CCZ4 [21] with damping $(\kappa_1, \kappa_2, \kappa_3) = (3, 0, 1)$, 1+ log slicing [35] and a Gamma-driver shift [36, 37] with $\eta = 1$, fourth-order stencils, RK4, Kreiss–Oliger dissipation $\sigma = 2$, $\Delta t = 0.02\,\Delta x$, and Sommerfeld outer boundaries. Every cell is *strictly uniform-grid* (`max_level` $= 0$): mesh refinement is disabled outright, so no tagging criterion or refinement boundary enters any number here, and the resolution ladder below means exactly one thing—the grid spacing.

The main series (`convA`) uses a domain $L = 64$ at $N = 128$, 192, 256, i.e. $\Delta x = 0.50, 0.33, 0.25$—about 10, 15 and 20 points across a stellar rms radius of ≈5. Every configuration is run at all three; Weyl shells sit at $R = 8$ and 16. A second series (`boxC`) doubles the domain to $L = 128$ at $\Delta x = 0.50$ with shells at $r = 16$, 24, 32, 40, for the wave-zone test of Sec. VI B, where the light-crossing time to the boundary rather than the resolution is what matters. Stars are released *at rest*—no boosts, no driving—so any drift is gravitational.

The release separation is where the loudest signal and the cleanest measurement disagree. The dressed stars have rms radius ≈5 and $R_{90} \approx 7.6$, so at $d_0 = 8$ the envelopes overlap from $t = 0$ and by $t = 60$ the pair has closed to a gap of 1.86: unmistakable, and the best pictures, but late times measure a merger rather than a force law. The reference configuration for every quantitative statement is therefore $d_0 = 12$ at $\Delta x = 0.25$, where the stars start more than R_{90} apart and the gap closes only from 12.00 to 10.14. The $d_0 = 8$ cells are retained as the strong-signal illustration, labelled contact-contaminated wherever they appear; a third separation, $d_0 = 16$, fails the convergence test of Sec. V C and is quoted as a bound, never as a measurement.

B. Diagnostics

The primary diagnostic is the per-sector barycentre stream, sampled continuously: for each sector $s = \pm$,

$$\bar{x}_i^{(s)}(t) = \frac{\int_{\mathcal{D}} w_s\, x_i\, d^3x}{\int_{\mathcal{D}} w_s\, d^3x}, \qquad w_s = \left(|\Phi_s|^2 + |\Pi_s|^2\right)^{1/2}, \quad (8)$$

together with the total weight $\int w_s\, d^3x$ and the rms radius about $\bar{x}^{(s)}$. The accounting region $\mathcal{D}$ is the full computational domain in coordinate volume; its finite extent, and the coordinate nature of the centroid, are the source of the biases quantified in Sec. VII.

The headline quantity is the *pair barycentre* $X = \frac{1}{2}(\bar{x}^{(+)} + \bar{x}^{(-)})$ and its drift ΔX: it is what Bondi's argument predicts to accelerate, it is defined identically for pairs and controls, and by symmetry it is exactly zero for every control in the matrix—so a nonzero control drift is pure systematics, making Sec. V B a measurement of the noise floor rather than a guess at it. Constraint norms (Appendix A), confinement, the $\ell = 2$ modes of Ψ_4, and energy-condition monitors are recorded alongside.

Table I is the experiment: each row is a falsifiable prediction of the sign structure—a same-sector pair or lone star that drifted like the mixed pair would kill the claim. `PM-eq` changes exactly one physical quantity (the phantom's mass, by 17%); `MP` reflects the geometry; the separations probe the force law.

TABLE I. The falsifiable run matrix. The sign structure of Sec. II predicts a drifting pair barycentre for the mixed pair, reversal under mirroring, a weaker runaway at wider separation, and *no* net drift for same-sector pairs or lone stars. Drifts in code units at $t = 60$ ($t = 40$ for the singles). Unless marked, values are the finest campaign grid, $\Delta x = 0.25$; the mirror and single-star cells exist only at the production resolution $\Delta x = 0.5$ and are marked †. Every cell was solved to Ham ≤0.095%, Mom ≤0.080% (Appendix A).

run	$\Delta x^{(+)}$	$\Delta x^{(-)}$	ΔX	verdict
`PM`, d_0=12 (ref.)	+0.71	+2.57	**+1.64**	**runaway**
`PM`, d_0=8[a]	+3.33	+9.48	+6.40	runaway
`PM-eq`, d_0=8[a]	+2.94	+9.51	+6.22	runaway
`PM`, d_0=16[b]	−0.33	+0.59	+0.13	bound only
`MP` mirror†	−3.23	−9.07	−6.15	reversed
`PP` $(+,+)$	+0.036	—	+0.036	null
`MM` $(-,-)$	—	+0.012	+0.012	null
`P` (single +)†	+0.061	—	+0.061	null
`M` (single −)†	—	+0.065	+0.065	null

[a] Contact-contaminated: the envelopes overlap from $t = 0$ and the gap closes to 1.86 (1.42 for `PM-eq`) by $t = 60$. Illustration, not a force-law measurement.

[b] Not grid-stable: the drift spreads 53% across the resolution ladder (Sec. V C), and the canonical column is additionally wrong-signed because the phantom's repulsion pushes canonical matter out through the $+x$ boundary (Sec. VII).

V. RESULTS

A. The runaway

In the reference cell—canonical star released at rest at $x = +6$, phantom at $x = -6$—*both* barycentres move toward $+x$ (Figs. 3 and 4). By $t = 60$ the canonical star has moved +0.71 and the phantom +2.57, so the pair barycentre has drifted $\Delta X = +1.64$ while the gap closed from 12.00 to 10.14. The motion is a sustained acceleration from rest, not an impulse: ΔX passes +0.018 at $t = 15$, +0.033 at $t = 20$, +0.131 at $t = 30$ and +0.380 at $t = 40$. The barycentre *speeds* say the same thing (code units, $c = 1$): the phantom reaches $v_x \approx 0.027$ at the clean-window edge $t = 30$ and ≈ 0.15 by $t = 60$, the canonical star ≈ 0.004 and a peak near 0.047 at $t \approx 52$ before the halo bias of Sec. VII drags the tracked centroid

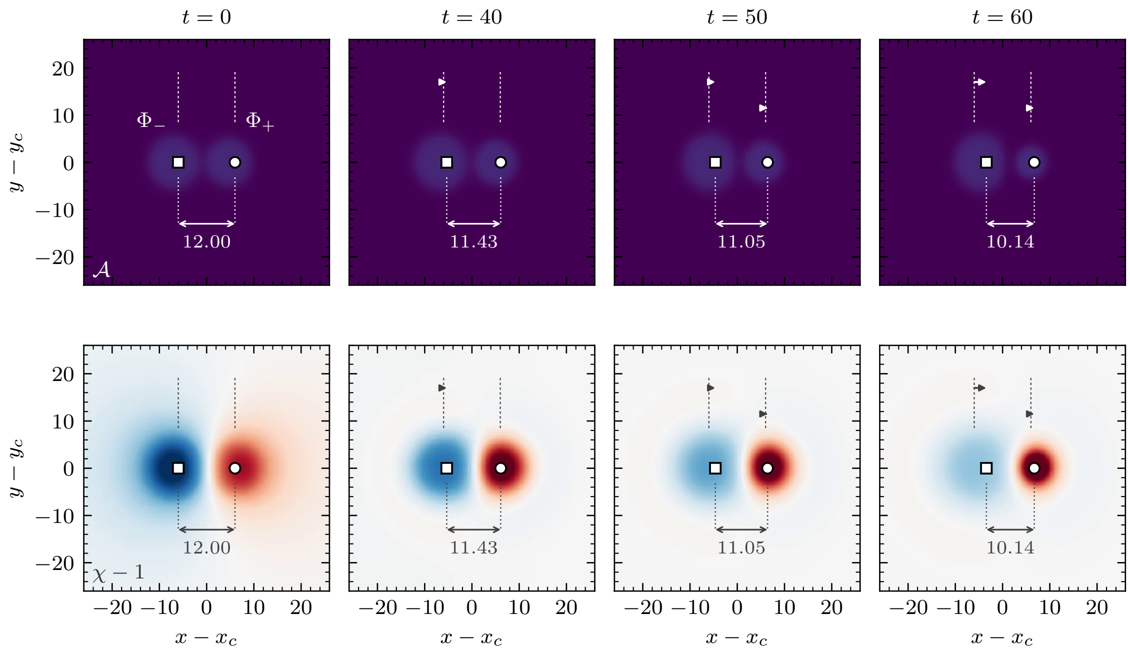

FIG. 3. The runaway in the reference cell (`PM`, $d_0 = 12$, $\Delta x = 0.25$), $z = 0$ plane, at $t = 0$, 40, 50, 60. Top: scalar-field activity $\mathcal{A} = \sum_k (\varphi_k^2 + \Pi_k^2)^{1/2}$, both sectors. Bottom: $\chi - 1$; the phantom star is a hill ($\chi > 1$), the canonical star a well ($\chi < 1$). Open markers: measured sector barycentres (circle Φ_+, square Φ_-); dashed ticks mark the release positions $x = \pm 6$, arrows the displacement since release, and the bracketed number is the measured gap, $12.00 \to 11.43 \to 11.05 \to 10.14$. Both stars move toward $+x$ while the gap slowly closes. Panels are individually colour-scaled.

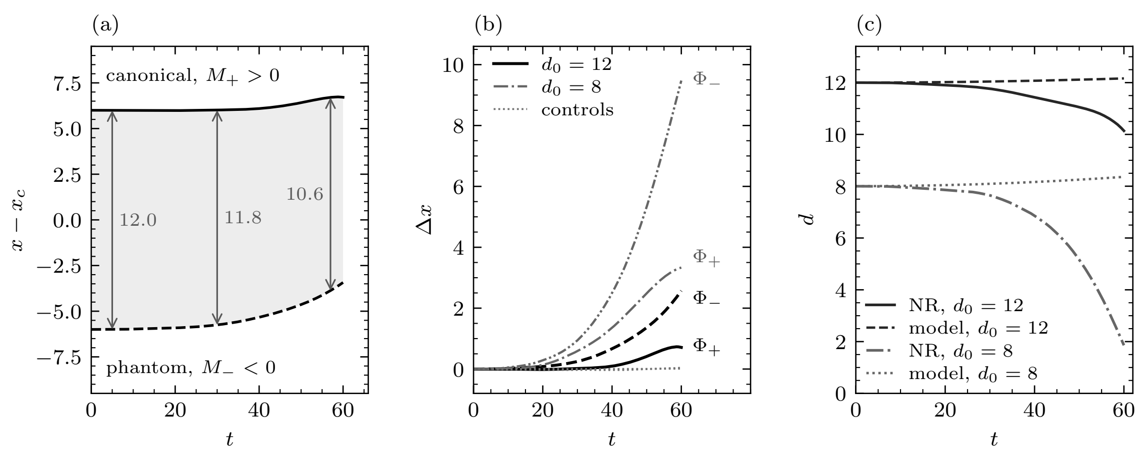

FIG. 4. Worldlines and drifts, finest campaign grids only. (a) The two sector barycentres of the reference cell, released at rest at $x = \pm 6$: both move toward $+x$, the phantom (dashed) chasing the canonical star (solid), with the gap at $t = 5$, 30 and 57 ($12.0 \to 11.8 \to 10.6$). (b) Sector drifts Δx: the reference $d_0 = 12$ cell in black, the contact-contaminated $d_0 = 8$ cell in gray, and the same-sign controls dotted along the axis; curve ends are labelled by sector. (c) The gap $d(t)$ for each configuration against its *own* matched point-mass Bondi integration: the model keeps the gap open, the simulations close it, and the discrepancy shrinks with separation (Sec. V E).

back. In the contact-contaminated $d_0 = 8$ cell the same curves run several times higher, which is why that cell makes the better picture and the worse measurement.

Two features of the reference run are diagnostic artefacts rather than physics, both quantified in Sec. VII: the canonical sector's drift is slightly *negative* over $12 \lesssim t \lesssim 25$ (worst -0.016 near $t = 19$), and its tracked weight decays as shed field exits the domain. Neither affects the phantom sector, and neither can produce a systematic one-signed drift of the pair barycentre—which is why the controls below, not an error model, set the noise floor.

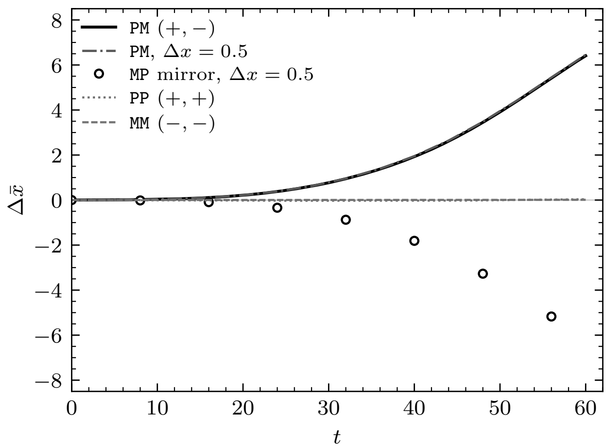

FIG. 5. The four sign assignments under identical numerics: pair-barycentre drift at $d_0 = 8$. The mixed pair PM runs away to $+6.41$ at $\Delta x = 0.25$ and $+6.45$ at $\Delta x = 0.5$, its space-reflected twin MP to -6.15 at the same $\Delta x = 0.5$ (a 4.7% asymmetry), while PP and MM stay on the axis: $+0.036$ and $+0.012$ at $t = 60$, 179 and 545 times smaller than the mixed pair, below one cell width of the finest grid, and shrinking under refinement (Fig. 6b).

B. Only the mixed pair moves

The controls are the null hypothesis, run under numerics identical to the pairs—same code, gauge, dissipation, solve tolerance and grids (Fig. 5). At $t = 60$ the two-canonical pair PP has drifted $+0.036$ and the two-phantom pair MM $+0.012$, against the reference pair's $+1.64$: factors of 46 and 142. Both residuals are smaller than one cell width of the grid they were measured on, and both *shrink* as that grid is refined—PP runs $+0.149$, $+0.080$, $+0.036$ down the ladder and MM $+0.039$, $+0.014$, $+0.012$—the signature of a numerical residual, not a physical drift. The single stars, available at $\Delta x = 0.5$, drift $+0.061$ and $+0.065$ by $t = 40$. The same-sign controls sit at $d_0 = 8$ rather than at the reference separation, because they were built alongside the original $d_0 = 8$ series; a closer pair is if anything *more* prone to spurious centroid motion, so using them as the floor is conservative—against the $d_0 = 8$ mixed cell the same controls come out 179 and 545 times smaller.

A drift produced by grid anisotropy, boundary asymmetry or diagnostic bias would not know which star is which; swapping the sectors in place must reverse the physics but not an artifact. It reverses the physics. Reflected about $x = 0$, the mirrored run MP reproduces the reference trajectory of the same numerics to 3.4% in the canonical sector and 5.2% in the phantom sector at $t = 60$, and its pair barycentre reaches -6.15 against $+6.45$ for the unmirrored run—a 4.7% asymmetry. Both runs exist only at $\Delta x = 0.5$, which is the correct control anyway: a mirror test must hold everything except the reflection fixed.

C. Grid convergence

Every configuration was run at $\Delta x = 0.50$, 0.33 and 0.25 under otherwise identical settings. The logic is the one that applies whenever an effect and its possible artefacts live on the same grid: a physical drift should be insensitive to the grid, a numerical one should converge away. Figure 6 shows both halves happening at once.

In the reference cell the three curves lie on top of one another for the whole run, ending at $+1.746$, $+1.677$ and $+1.642$—a spread of 6.3% at $t = 60$ and 6.6% at $t = 30$. Early in the run the same absolute scatter is a larger fraction of a much smaller signal (12.9% at $t = 20$, where ΔX is only $+0.033$); that is the error bar carried by the earliest entries of Table II, and why the ratio tests there are built from same-resolution run pairs, in which it largely cancels. The contact-contaminated $d_0 = 8$ cell and its equal-$|M|$ variant are tighter throughout (0.5% and 0.3% at $t = 60$) simply because the signal is larger against the same noise. Meanwhile the PP residual falls by roughly two at each refinement step and MM by three overall, so the gap between signal and floor *widens* with resolution: refinement does not erode the runaway, it isolates it.

At $d_0 = 16$ the test fails, and we report it as such: the drift spreads 17% at $t = 15$, 41% at $t = 30$ and 53% at $t = 60$—the signal has fallen to the level where the grid still controls it. Every statement about $d_0 = 16$ here is a bound.

One caveat must accompany all of this: the campaign cannot report a convergence *order*, because a single elliptic solve on its own fixed grid supplies the initial data for all three evolutions of a ladder, and its residual dominates the constraint violation from $t \gtrsim 1$ onward (Appendix A). Richardson extrapolation against that floor would measure the solve, not the evolution; we quote the ladder spread as the empirical error bar and defer a formal order to a campaign in which the solve tolerance is refined together with the grid.

D. Gravity scales with the source

The strongest quantitative tests compare runs differing in exactly one quantity. Both are available in the $d_0 = 8$ family, where PM-eq retunes the phantom to $|M_-| = M_+$ and changes nothing else; the contact contamination affects both runs identically and largely cancels from the ratios.

The push scales with the phantom's active mass. Between PM and PM-eq the phantom's mass changes by the factor $0.06395/0.07696 = 0.831$, and the canonical star's displacement responds in proportion (Table II): the measured drift ratio is 0.825 at $t = 15$, 0.843 at $t = 20$ and 0.863 at the window edge $t = 30$—within 0.7%, 1.5% and 3.9% of the bare mass ratio, and within 0.8–3.5% of the matched point-mass ratio, which drifts the same way and for the same reason (rising to 4.9% at $t = 40$, as the closing gap begins to matter).

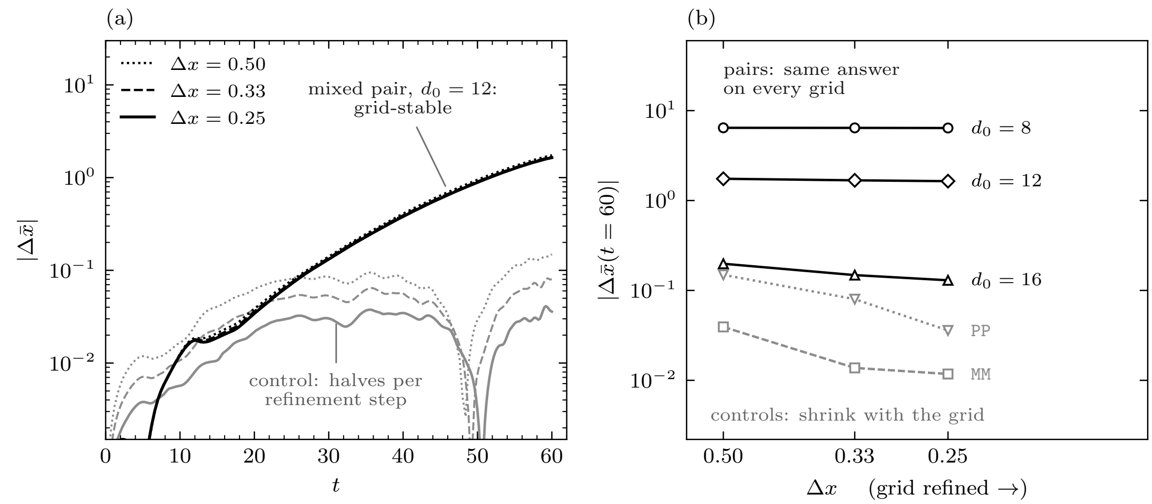

FIG. 6. The convergence study. (a) The reference cell's resolution ladder in time (dotted $\Delta x = 0.50$, dashed 0.33, solid 0.25) against the PP control's ladder: the mixed-pair curves lie on top of one another while the control's residual drift roughly halves at every refinement step. (b) The same data as a refinement sequence—$|\Delta\bar{x}|$ at $t = 60$ against grid spacing, grid refined to the right—for every family. The mixed pairs sit on plateaus (spread 0.5% at $d_0 = 8$, 6.3% at $d_0 = 12$) while the same-sign controls fall towards zero. The under-resolved $d_0 = 16$ family is the exception that shows the test works: its 53% spread is why it is quoted as a bound.

TABLE II. Does the push scale with its source? Between `PM-eq` and `PM` one physical input changes: the phantom's mass magnitude, smaller by the factor 0.831. "NR ratio": the canonical-star drift of `PM-eq` over that of `PM`, both at $\Delta x = 0.25$; beside it the same ratio from the matched point-mass integrations, and the bare mass ratio. The last column is the phantom's own drift ratio, which the sign structure requires to be 1.

t	NR ratio	point-mass	mass ratio	Φ_- ratio
15	**0.825**	0.832	0.831	1.022
20	**0.843**	0.832	0.831	1.004
30	0.863	0.834	0.831	1.025
40	0.877	0.837	0.831	1.033
60	0.881	0.843	0.831	1.004

The fall depends only on the other body. In the same pair of runs the canonical source is unchanged, and so is the phantom's trajectory: drift ratio 1.022 at $t = 15$, 1.004 at $t = 20$, 1.025 at $t = 30$ and 1.004 at $t = 60$. Two independent solves and evolutions, differing in the phantom's own mass by 17%, reproduce its trajectory to 0.4–2.5% across the clean window—a physics statement and a reproducibility check at once.

E. Separation dependence

The remaining knob is the release separation, and it settles what the excess over Newtonian point masses is made of. Fixed-time power-law fits are useless here—the closing gap makes them report exponents between 2 and 5, measuring runaway feedback rather than the force law—so each run is compared against its *own* point-mass Bondi integration: two point particles carrying the stars' ADM masses, released at rest at the same separation, each accelerated by the other's active mass (Table III).

The pattern is unambiguous. At $d_0 = 8$ the phantom's fall runs 1.4× the point-mass rate at $t = 20$ and 5.4× by $t = 60$: it sits deep inside the canonical star's near field, where the local pull exceeds M/d^2 evaluated at the barycentric separation, and the canonical well deepens further as it re-accretes its shed field ($\chi_{\min}$ falling from 0.98 to 0.43). At the reference separation the excess is much smaller—0.93× at $t = 20$, 1.24× at $t = 30$—so the measured motion is Newtonian to within 7% early and 24% at the window edge, the excess growing only as the gap starts to close in earnest (12.00 → 10.14 by $t = 60$, against 12.16 for the point-mass model). At $d_0 = 16$ the ratios are 0.87 and 0.98, consistent with the point-mass law, though there the drift is not grid-stable and the agreement is only a bound.

F. The phantom sector is the stable one

Counter to intuition, the phantom star was the best-behaved object in the campaign. Gravity is not its binding agent—the scalar self-interaction is—and the reversed sign is the *stabilizing* correction: shed field around the canonical star is pulled back, deepening the well and feeding perturbations, while around the phantom it anti-gravitates and self-dilutes. The extremal conformal factor makes the contrast quantitative at $\Delta x = 0.25$: by $t = 60$ the reference cell has reached $\chi_{\min} = 0.205$,

TABLE III. The runaway against matched point-mass Bondi integrations: two point particles carrying the stars' ADM masses, released at rest at the same separation, each accelerated by the *other's* active mass. Left: the phantom's drift in the reference cell as measured and as modelled. Right: measured over modelled phantom drift for each separation, all at $\Delta x = 0.25$. The phantom sector carries the comparison because the canonical centroid is biased low by the halo (Sec. VII); the clean window is $t \lesssim 30$, and the $d_0 = 16$ entries bound rather than measure (Sec. V C).

	Φ_- drift, d_0=12		NR/model by d_0		
t	NR	model	8	12	16
20	+0.08	+0.09	1.43	0.93	0.87
30	+0.25	+0.20	2.10	1.24	0.98
40	+0.67	+0.35	3.14	1.88	1.31
60	+2.57	+0.80	5.35	3.23	1.32

the $d_0 = 8$ cell 0.434 and the merging `PP` control 0.499, while the mutually repelling `MM` control sits at 0.989 after passing through 1.002—flat to a part in 10^3 while its material spreads. The single-star records agree: over $t = 0 \to 40$ at $\Delta x = 0.5$ the canonical star's extremal χ falls 0.989 $\to$ 0.851 and its tracked rms radius more than doubles, 5.05 $\to$ 12.25, with $\pm 8\%$ peak breathing, while the phantom's χ moves only 1.000 $\to$ 1.001, its rms radius grows by a seventh, 5.43 $\to$ 6.25, and it breathes at $\pm 3\%$.

VI. THE GRAVITATIONAL-WAVE SIGNATURE

A uniquely relativistic consequence of the runaway is gravitational bremsstrahlung: the exact accelerating-pair metrics are radiative [2, 3], and negative-mass binaries have been proposed as sources of anomalous signatures with no positive-mass counterpart [38]. Our runs record the five $\ell = 2$ spin-weighted amplitudes $r\,\psi_4^{2m}$ of the Weyl scalar Ψ_4, summarized by $A_{\ell=2} = [\sum_m |r\,\psi_4^{2m}|^2]^{1/2}$. Near zone and wave zone answer different questions, and only the second needs the larger box.

A. The near-zone signal tracks the source

On the $L = 64$ domain both shells ($R = 8$ and 16) sit inside the massive-field radiation bath, and the amplitude falls steeply between them—$A(R{=}8)$ exceeds $A(R{=}16)$ by 5× at $t = 30$ and 10× at $t = 60$, far from the r^0 peeling of a wave-zone signal. What follows is therefore near-zone curvature: the geometry's independent testimony about the source, not a radiated flux.

That testimony is unambiguous (Fig. 7). The waveform resembles no binary signal in the astrophysical catalogue—a monotone, chirpless ramp, no oscillatory train, no propagating front. In the mixed cells, and only there, $A_{\ell=2}$ climbs smoothly, ending 122 times above its

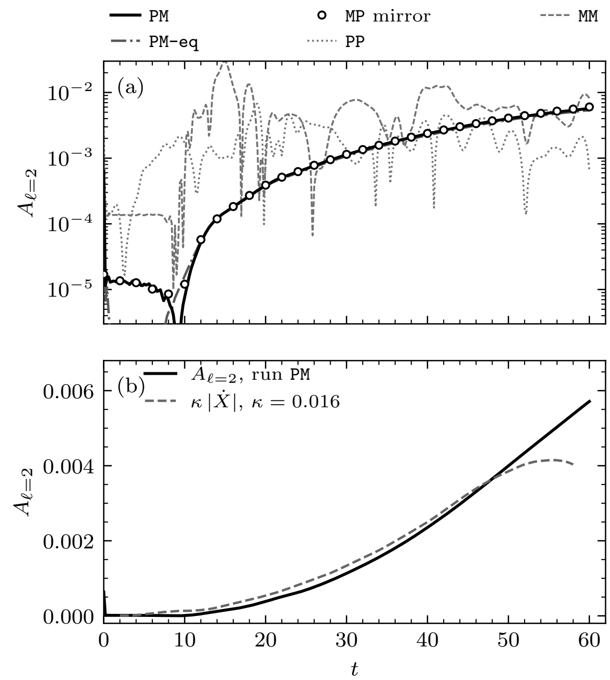

FIG. 7. The $\ell = 2$ Weyl signal at $R = 16$, campaign cells at $\Delta x = 0.25$ (the mirror overlay exists only at $\Delta x = 0.5$). (a) Total amplitude $A_{\ell=2}(t)$: the mixed pairs climb by two orders of magnitude (232× for `PM`, 350× for the equal-$|M|$ rerun, 481× for the mirror, 122× in the reference $d_0 = 12$ cell) while the controls fluctuate at their bath level. (b) The amplitude against the rescaled pair speed $\kappa|\dot{X}|$: correlation 0.996 over $5 \le t \le 55$ with $\kappa \approx 0.016$—the near-zone quadrupole rises in lockstep with the runaway, not with the field frequency.

early-time floor in the reference cell and 232–481 times in the louder $d_0 = 8$ family; the same-sign controls fluctuate at their bath level. The growth is locked to the runaway kinematics—over $5 \le t \le 55$ the amplitude correlates with the pair speed $|\dot{X}|$ at 0.996, with a single scale factor $\kappa = 0.0163$—as the source structure dictates, since the signed quadrupole of a rigidly translating dipole grows with the drift, $Q \sim 2M d\, X(t)$. Three checks tie the signal to the physics rather than the bath or the grid: power resides in $m = 0, \pm 2$ with $m = \pm 1$ at 0.54%, the pattern an x-axisymmetric source imprints on a z-axis decomposition; 99.7% of the spectral power lies below $\omega = 0.4$, with 3.8×10^{-4} at the field frequency and 7.5×10^{-6} at 2ω, so the $\ell = 2$ curvature is not leakage from the oscillating scalar; and the mirrored run reproduces the amplitude history to 4.7%, extending the artifact rejection of Sec. V B from matter to curvature.

B. One outgoing wave in the wave zone

To ask whether any of this is radiation, the shells have to leave the bath, which requires the box rather than the resolution. The `boxC` series doubles the domain to

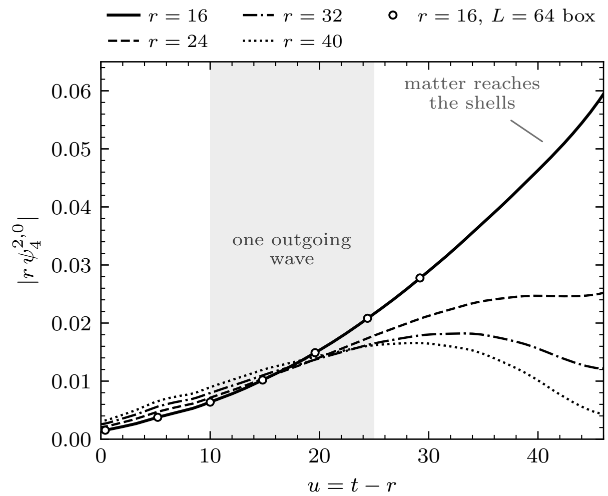

FIG. 8. One outgoing wave, from the double-size box ($L = 128$, $\Delta x = 0.5$). $|r\,\psi_4^{2,0}|$ on the four wave-zone shells $r = 16$, 24, 32, 40, each against its own retarded time $u = t - r$. Inside the shaded window $u = 10$–25 the shells agree to 1.10–1.40×; before it they carry the start-up transient, after it the expanding matter reaches them. Open circles: the half-size $L = 64$ box at $r = 16$, differing by 0.07–0.18%—the amplitude is box-independent.

$L = 128$ and places four shells at $r = 16$, 24, 32 and 40. One outgoing wave gives the same $|r\,\psi_4^{2,0}|$ on every shell when each is read at its own retarded time $u = t - r$; Fig. 8 performs that test.

There is a clean window, bounded on both sides for understood reasons. Inside $u = 10$–25 the four shells agree to 1.40×, 1.16×, 1.10× and 1.33×—one wave, propagating outward at the speed of light, on shells spanning a factor of 2.5 in radius. Before $u \approx 10$ they carry the initial-data start-up transient; after $u \approx 25$ the expanding canonical envelope reaches them (its rms radius crosses $r = 16$ at $t \approx 51$) and they decouple, spreading to 4.6× by $u = 40$. Doubling the box changes the amplitude on the common $r = 16$ shell by 0.07–0.18% across that window, so the signal is not a boundary reflection.

What this does *not* settle is the energy budget—whether the net energy radiated is positive, negative, or cancels between the sectors. That needs the odd-ℓ modes, ADM surface integrals at the outer boundary, and a sponge layer to extend the window past the arrival of the matter: the geometry-side program for the follow-up campaign, benchmarked against the Bonnor–Swaminarayan acceleration parameter [2].

C. Is there dipole radiation?

For binaries whose members carry different ratios of gravitational charge to inertial mass the leading anomalous channel is dipolar [38]. The bicomplex model closes it structurally: the source of gravity is the signed stress tensor, conserved on shell (Sec. II), so the rate of change of the signed mass dipole is the conserved signed momentum—near zero for pairs released at rest—and mass-dipole radiation is forbidden at leading order for exactly the reason it is forbidden in ordinary general relativity. Kinematics agrees from the other side: Ψ_4 carries spin weight -2, so its multipole expansion begins at $\ell = 2$ and there is no $\ell = 1$ mode to populate whatever the dipole does. The anomalous channel of a Bondi dipole is therefore not dipolar emission but the signature Fig. 7 shows in embryo: the ever-growing quadrupole of a linearly accelerating, near-zero-total-mass system—bremsstrahlung without inspiral, secular where every astrophysical binary chirps.

The gap closure of Sec. V E does not reinstate the channel. Dynamically, $\dot{D}$ is a momentum rather than a source, and which momentum matters: for the whole system it is the ADM momentum, conserved and zero for data released at rest, so the total $\ddot{D}$ vanishes identically; for the matter alone it is the volume-integrated sector momentum, which is *not* separately conserved. The stream measures exactly that distinction: in `PM-eq`, where $M_+ + M_- = 0$ reduces the matter dipole rate to $M_+\dot{d}$, the measured signed momentum tracks $M_+\dot{d}$ to 6–14% across the clean window. The matter dipole does accelerate; the compensating momentum resides in the gravitational field, which is why closing the balance requires surface integrals rather than volume ones. What the closing gap genuinely adds is a second time dependence in the *quadrupole*, superposed on the linear acceleration that dominates Fig. 7.

One structural feature of the budget does not wait on the measurement: it has no floor. The pair begins with essentially no mass to spend—$M_+ + M_- = -0.013$ in the reference configuration, zero to five digits in `PM-eq`—while the news-squared flux is positive-definite irrespective of what sources the field, so a radiating isolated system loses mass monotonically, and this one starts from zero. Nothing arrests the descent: the bound that would is the positive-mass theorem, whose hypothesis this matter violates by construction. The runaway is secular in its energetics as much as in its kinematics—the classical face of the vacuum instability that already bars this matter quantum-mechanically [31, 32]—and it sharpens the wave-zone program: the question is not only how large the flux is, but whether its sign is the positive one this argument presumes.

VII. CAVEATS, AND THEIR MEASURED SCOPE

The barycentre includes the halo, and biases the canonical centroid low. Equation (8) integrates core plus shed radiation over the whole domain. In the reference cell the canonical sector's tracked weight holds to 4.2% through $t = 30$, then falls to 0.39 of its initial value by $t = 60$ as field leaves the accounting region; the phantom's instead grows (1.22× by $t = 60$). Losing the leading edge of a distribution drags its centroid backwards, so canonical drifts are underestimates everywhere, worst where the true displacement is small: the reference run reads slightly negative over $12 \lesssim t \lesssim 25$, and at $d_0 = 16$ the canonical sector never recovers, ending at −0.33. The clean quantitative window is therefore $t \lesssim 30$, the phantom sector is the trustworthy one and carries Sec. V E, and the mass-scaling ratios compare equal-separation runs so the bias largely cancels there. A core-weighted centroid immune to the halo is built for the follow-up.

Resolution. The ladder establishes that the effect is not a grid artefact, not a convergence order (Sec. V C); refining the solve alongside the grid is the outstanding piece of work. The test also has a floor of its own—at $d_0 = 16$ it fails—which is why that separation is a bound throughout.

The radiation bath cannot leave. Massive-field radiation reflects from massless-wave (Sommerfeld) boundaries and washes back over the stars; this caps the mixed runs at $t = 60$. Doubling the box shows the extracted amplitude is boundary-insensitive at the 0.2% level inside the clean window (Sec. VI B), but does not remove the bath; a sponge layer is the fix, and is what a constant-gap test at $t \gg 60$ requires.

Gauge. The barycentres of Eq. (8) are coordinate quantities in a dynamical gauge. The mirror test and the Weyl correlation are the two checks that do not depend on that choice, and both agree with the coordinate measurement.

VIII. DISCUSSION AND CONCLUSION

We posed Bondi's positive–negative mass pair as a full initial-value problem, and it behaved as he argued in 1957.

(1) The runaway is real in full general relativity. Released at rest, the mixed pair—and only the mixed pair—self-accelerates: the pair barycentre drifts +1.64 by $t = 60$, 46 and 142 times the same-sign controls, and reverses under mirror reflection to 4.7%, with the $\ell = 2$ Weyl amplitude rising in lockstep with the runaway speed (correlation 0.996).

(2) The measurement survives grid refinement. The reference drift is stable to 6.3% across the ladder while the control residuals fall by factors of 3–4: refinement widens the separation between signal and floor. No convergence order is quoted, and the reason—a shared elliptic-solve floor—is stated rather than hidden.

(3) The effect responds to its source the way gravity must. The push on the canonical star scales with the phantom's active mass to 0.7–3.9%; the phantom's fall is set by the canonical star alone, reproducing across independent solves to 0.4–2.5%; and the excess over the matched point-mass prediction falls monotonically with separation (2.10×, 1.24×, 0.98× at $t = 30$).

(4) The momentum bookkeeping balances. The two sectors' volume-integrated signed momenta—each growing to (5–7) × 10^{-3} by $t = 30$—cancel to a residual 2.6–4.0× smaller, and that residual is not truncation error: it reproduces the point-mass sum $M_+v_+ + M_-v_-$, built from independently tracked core velocities and the dressed ADM masses, to about 10% over $20 \le t \le 30$. A pair of net mass −0.013 *must* carry $P = M_{\rm net}v$; the field-side closure—matter plus gravitational-field momentum summing to zero—awaits the ADM surface integral on an enlarged domain.

(5) A stable star with negative ADM mass exists as constraint-satisfying, asymptotically flat initial data and survives dynamical evolution—to our knowledge a first in this setting (static phantom-field stars [30] and de Sitter bubbles [28] were known), and possibly the campaign's most reusable artifact.

What is not established. A *constant-gap* runaway—the textbook pair accelerating forever at fixed separation—requires point-like separation, and we do not reach it: at $d_0 = 8$ the stars overlap, at the reference separation the closure is mild but present, and at $d_0 = 16$ the gap is nearly constant but the drift is not grid-stable. The clean test needs wide separation *and* a longer window—a sponge layer and a larger domain, not a finer grid. Raising ω helps only partially: the canonical star's R_{90} shrinks by a bounded 32% toward $\omega \simeq 0.80$ and the sector mass asymmetry improves (20.3% → 4.3%; Fig. 2), but the compactness $2|M|/R_{99}$ *falls* 6.4× over the same interval, so higher ω buys a better separation-to-size ratio at the cost of a weaker signal and cannot substitute for the larger domain.

Every mixed run was stopped before contact; what a canonical–phantom collision produces—including whether a horizon forms [11]—is untouched. And nothing here bears on whether NEC-violating matter exists. The contrast with Nojiri and Odintsov [12] is instructive: their positive–negative pair, immersed in a cosmological fluid with a negative cosmological constant, can bind; ours is asymptotically flat and isolated, with nothing to bind against, and it runs away. The fate of a Bondi dipole is a statement about its environment as much as about its signs.

Next steps. A core-weighted centroid, and a solve tolerance refined together with the grid so that a convergence order becomes quotable; a sponge layer to unlock $t \gg 60$ and the constant-gap test at $d_0 = 16$; wave-zone extraction on the enlarged domain—odd-ℓ modes, ADM surface integrals, the Bonnor–Swaminarayan benchmark [2]—to settle the sign and budget of the energy this runaway radiates. The contact end state follows.

REPRODUCIBILITY

All time series, star tables, parameter files, code patches, analysis scripts and the campaign journal behind every number and figure are packed, with machine paths scrubbed, in the repository's results tree (`results/bondi-dipole-runaway/`); a single script regenerates the figures and prints every number quoted in the captions and text. Every run lives in campaign/, one directory per cell (convA_<config>_n<N> for the $L = 64$ convergence series, boxC_<config>_L128_n256 for the double box; the reference cell is convA_pm_sep12_n256). Four earlier production runs at $\Delta x = 0.5$ sit alongside under unprefixed names, used only where the convergence campaign has no twin. Initial data: GRTresna [24]; evolution: GRTeclyn [22, 23, 25].

ACKNOWLEDGMENTS

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Appendix A: Constraint behaviour

Initial data. GRTresna's nonlinear iteration contracts monotonically on every cell: the mixed-pair solves converge in seven iterations to 0.083% (Hamiltonian) and 0.075% (momentum), and the worst cell in the campaign exits at 0.094%/0.079%—the gate quoted in Table I, in the solver's own grid and normalization.

Evolution. The constraints are re-evaluated on the evolution grid every $\Delta t = 0.5$ and reduced to composite norms over the domain (outermost four cells excluded, so boundary violation is not misread as interior error). Figure 9 shows the reference cell's full ladder in absolute code units: over an overwhelmingly vacuum domain a global *relative* normalization is dominated by near-zero denominators, so the yardstick is the controls run under identical numerics. On the finest grid $L_2(\mathcal{H})$ starts at 2.9×10^{-3}, relaxes within $t \lesssim 1$ to the truncation floor 1.0×10^{-4}, then grows to 1.2×10^{-3} at $t = 15$, 2.8×10^{-3} at the window edge $t = 30$ and 7.7×10^{-3} at $t = 60$ (L_∞: 5.5×10^{-2} and 1.5, concentrated in the deepening canonical well as the trapped bath washes back); $L_2(\mathcal{M})$ stays below 3.4×10^{-4} through $t = 45$. This growth is the quantitative face of the $t \lesssim 30$ window. Through that window the mixed run's violation is within a factor 1.8 of the non-drifting controls while the drifts differ by 46–142×: the drift correlates with the sign structure, not the constraint error.

And the ladder itself, which is the point of Sec. V C's caveat: refining the grid does *not* reduce the violation. The coarse-to-fine ratio of $L_2(\mathcal{H})$ is 1.04–1.06× at every sampled time in every mixed-pair ladder, where a truncation-dominated violation would fall by ∼16 between $\Delta x = 0.50$ and 0.25 at fourth order. It does not, because all three evolutions inherit the same initial data from one elliptic solve on one fixed solver grid; that residual sets the floor, the evolution grid never gets below it, and that is why the campaign quotes ladder spread as an error bar and no convergence order.

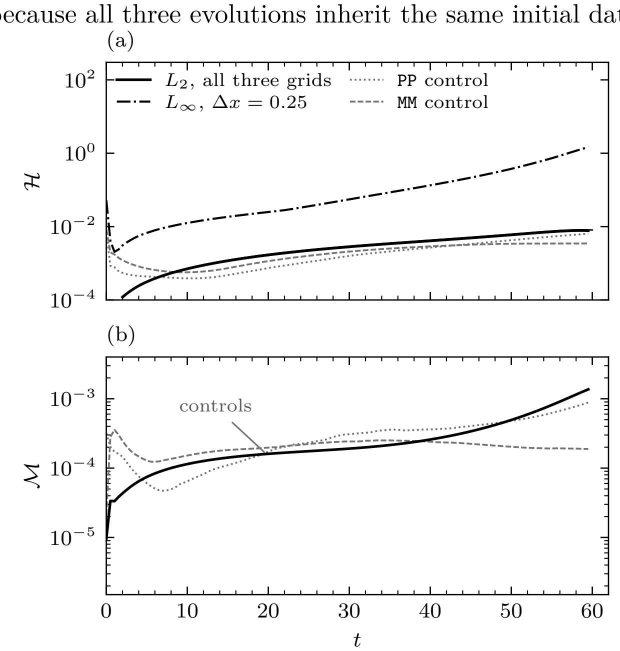

FIG. 9. Constraint monitoring for the reference cell's resolution ladder (dotted $\Delta x = 0.50$, dashed 0.33, solid 0.25), against the same-sign controls under identical numerics. (a) Hamiltonian constraint $\mathcal{H}$: composite L_2 norms, plus the finest grid's L_∞. The three ladder curves coincide—the elliptic-solve floor discussed in the text. (b) Momentum constraint $\mathcal{M}$, L_2.

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